

Homework Grading Report

Student:	Student_10
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	15.0	Missing: %>%()
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Detailed Feedback

What You Did Well

- Your code consistently uses the pipe operator (%) effectively throughout multiple sections, making your data transformations readable and maintainable. For example, in Part 3 you chained filter, select, arrange, and slice_head operations cleanly.
- You correctly implemented data cleaning by using na.omit() to remove rows with missing values in sales_clean, which aligns with best practices for handling missing data in business analytics projects.
- Your approach to calculating outlier thresholds using IQR method is accurate and well-documented. You clearly defined Q1, Q3, IQR, and bounds before creating the outlier_analysis dataframe.
- Your use of group_by() and summarize() functions in Parts 4.3 and 4.4 demonstrates solid understanding of aggregation techniques for business intelligence analysis.

- You properly applied `pivot_wider()` to reshape data from long to wide format in Part 5, correctly specifying `names_from` and `values_from` arguments to create meaningful column names.
- Completed Part 1: R Basics and Data Import (10.0/10 points)

What to Fix

1. To improve your answer to Question 9.1, you need to include actual diagnostic output (e.g., `sum(is.na(sales_data$Revenue))` or boxplot visual) that confirms the absence of missing values or outliers. This matters because reviewers need proof that you examined data quality, not just an assumption. Add a code snippet that prints NA counts and uses IQR to flag outliers, then reference those numbers in your narrative.
2. For Question 9.2, you should cite specific figures such as "the West region generated \$1.2M in sales, 15% above the company average". Providing concrete numbers demonstrates that you can extract actionable insights from grouped summaries. Use `summarize(total_sales = sum(Total_Amount))` after `group_by(Region)` and quote the result.
3. Your response to Question 9.3 repeats the same example for both formats; you need to differentiate them. Explain that a wide format is ideal for a quarterly sales dashboard where each quarter is a column, while a long format is required for time-series regression where each row is a month-product observation. Show a small code example using `pivot_wider()` for the dashboard and `pivot_longer()` for the regression dataset.
4. In Question 9.5 you mention KPIs but do not specify which KPI you built. Identify a KPI you created (e.g., `Average_Order_Value = Total_Amount / n_items`) and explain why it matters for profitability analysis. Reference the `mutate()` line you used to calculate it and describe how the executive team could use that metric.

Reflection & Critical Thinking

In Question 9.1 you note that there were no missing values or outliers, then add a generic statement about why cleaning matters. The answer lacks concrete evidence from your dataset, such as a summary of NA counts or outlier detection, which would demonstrate that you actually inspected the data.

For Question 9.2 you correctly identify that grouped analysis reveals regional and category performance, but you do not provide specific metrics (e.g., top-selling region or category) that you observed. Including a numeric example would show you translated the summary into actionable insight.

Your response to Question 9.3 repeats the same sentences for both wide and long formats, resulting in redundancy and no clear distinction of when each format is optimal. A business scenario that requires a wide format (e.g., executive dashboard) versus a long format (e.g., time-series modeling) should be articulated separately.

In Question 9.4 you accurately describe the functional difference between `left_join` and `inner_join` and give appropriate business contexts. However, you could strengthen the answer by mentioning potential pitfalls, such as unintentionally dropping customers with no orders when using `inner_join`.

Question 9.5 correctly highlights Lesson 8 as valuable because it produces KPIs, yet you do not explain which KPI you created or why it matters for decision-making. Detailing a specific KPI (e.g., `Customer_Lifetime_Value`) would demonstrate deeper integration of the skill.

Instructor Assessment

You have covered all required sections of the exam and your code imports appear correct, which shows solid technical foundation. However, your reflection answers are too generic and lack the concrete evidence that demonstrates true mastery of the concepts. Strengthen each response by referencing actual numbers or code outputs from your analysis, and be explicit about how those findings would influence business decisions. Completing these improvements will raise both your analytical credibility and your overall grade.

Key Takeaway

Run diagnostic code to quantify missing values and outliers, then incorporate those statistics into your reflections.

When discussing grouped results, always include at least one numeric example to substantiate your insight.

Separate the business rationale for wide versus long data structures, and provide a brief code illustration for each.

Name specific KPIs you generate, explain their business relevance, and link them back to the case study objectives.